Algorithm 1 Advance level

DATA: Level l
Advance all grid at level l.
if $l < l_{max}$ **then**
  Apply **Algorithm 1** to level $l + 1$.
  Update all overlapping control volume at level l and correct flux for mass conservation.
  Regrid level l.
end if

Algorithm 2 Regrid level

DATA: Level l
Compute the local error using the error estimator on each grid at level l.
Create new grid at level $l + 1$ around the area where the local error is higher than the tolerance.
Prolongate the new grids at level $l + 1$
Copy the useful data from the previous grid at level $l + 1$ to the new grids at level $l + 1$
Reconnect the new grids at level $l + 1$ to the grids at level $l + 2$